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/*
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*/

#include <iostream>
using namespace std;

int price[100], n;

// Function to Input Product Prices
void input()
{
    cout << "Enter number of products: ";
    cin >> n;

    cout << "Enter Product Prices:\n";
    for (int i = 0; i < n; i++)
    {
        cin >> price[i];
    }
}

// Function to Display Product Prices
void display()
{
    cout << "\nProduct Prices are:\n";

    for (int i = 0; i < n; i++)
    {
        cout << price[i] << " ";
    }

    cout << endl;
}

// Bubble Sort
void bubbleSort(int sortOrder)
{
    for (int i = 0; i < n - 1; i++)
    {
        for (int j = 0; j < n - i - 1; j++)
        {
            if ((sortOrder == 1 && price[j] > price[j + 1]) ||
                (sortOrder == 2 && price[j] < price[j + 1]))
            {
                int temp = price[j];
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                price[j] = price[j + 1];
                price[j + 1] = temp;
            }
        }
    }
}

// Selection Sort
void selectionSort(int sortOrder)
{
    for (int i = 0; i < n - 1; i++)
    {
        int pos = i;

        for (int j = i + 1; j < n; j++)
        {
            if ((sortOrder == 1 && price[j] < price[pos]) ||
                (sortOrder == 2 && price[j] > price[pos]))
            {
                pos = j;
            }
        }

        int temp = price[i];
        price[i] = price[pos];
        price[pos] = temp;
    }
}

// Insertion Sort
void insertionSort(int sortOrder)
{
    for (int i = 1; i < n; i++)
    {
        int key = price[i];
        int j = i - 1;

        while (j >= 0 &&
              ((sortOrder == 1 && price[j] > key) ||
               (sortOrder == 2 && price[j] < key)))
        {
            price[j + 1] = price[j];
            j--;
        }

        price[j + 1] = key;
    }
}
```

```
// Partition Function for Quick Sort
int partition(int low, int high, int sortOrder)
{
    int pivot = price[high];
    int i = low - 1;

    for (int j = low; j < high; j++)
    {
        if ((sortOrder == 1 && price[j] <= pivot) ||
            (sortOrder == 2 && price[j] >= pivot))
        {
            i++;

            int temp = price[i];
            price[i] = price[j];
            price[j] = temp;
        }
    }

    int temp = price[i + 1];
    price[i + 1] = price[high];
    price[high] = temp;

    return i + 1;
}

// Quick Sort
void quickSort(int low, int high, int sortOrder)
{
    if (low < high)
    {
        int p = partition(low, high, sortOrder);

        quickSort(low, p - 1, sortOrder);
        quickSort(p + 1, high, sortOrder);
    }
}

// Main Function
int main()
{
    int choice, sortOrder;

    input();

    do
    {
```

```
cout << "\n===== PRODUCT PRICE SORTING SYSTEM =====\n";
cout << "1. Bubble Sort\n";
cout << "2. Selection Sort\n";
cout << "3. Insertion Sort\n";
cout << "4. Quick Sort\n";
cout << "5. Display\n";
cout << "6. Exit\n";

cout << "\nEnter your choice: ";
cin >> choice;

if (choice >= 1 && choice <= 4)
{
   cout << "\n1. Ascending Order\n";
   cout << "2. Descending Order\n";

   cout << "Enter your choice: ";
   cin >> sortOrder;
}

switch (choice)
{
   case 1:
      bubbleSort(sortOrder);
      cout << "\nBubble Sort Completed.\n";
      display();
      break;

   case 2:
      selectionSort(sortOrder);
      cout << "\nSelection Sort Completed.\n";
      display();
      break;

   case 3:
      insertionSort(sortOrder);
      cout << "\nInsertion Sort Completed.\n";
      display();
      break;

   case 4:
      quickSort(0, n - 1, sortOrder);
      cout << "\nQuick Sort Completed.\n";
      display();
      break;

   case 5:
      display();
```

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            break;

        case 6:
            cout << "\nThank You!\n";
            break;

        default:
            cout << "\nInvalid Choice!\n";
      }

   } while (choice != 6);

   return 0;
}
```

OUTPUT

```
Enter number of products: 5
Enter Product Prices:
500
200
800
300
100

=====PRODUCT PRICE SORTING SYSTEM=====
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Quick Sort
5. Display
6. Exit

Enter your choice: 5

Product Prices are:
500 200 800 300 100

=====PRODUCT PRICE SORTING SYSTEM=====
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Quick Sort
5. Display
6. Exit

Enter your choice: 1
```

1. Ascending Order
2. Descending Order

Enter your choice: 1

Bubble Sort Completed.

Product Prices are:
100 200 300 500 800

=====PRODUCT PRICE SORTING SYSTEM=====
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Quick Sort
5. Display
6. Exit

Enter your choice: 2

1. Ascending Order
2. Descending Order

Enter your choice: 2

Selection Sort Completed.

Product Prices are:
800 500 300 200 100

=====PRODUCT PRICE SORTING SYSTEM=====
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Quick Sort
5. Display
6. Exit

Enter your choice: 3

1. Ascending Order
2. Descending Order

Enter your choice: 1

Insertion Sort Completed.

Product Prices are:

100 200 300 500 800

=====PRODUCT PRICE SORTING SYSTEM=====
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Quick Sort
5. Display
6. Exit

Enter your choice: 4

1. Ascending Order
2. Descending Order

Enter your choice: 2

Quick Sort Completed.

Product Prices are:
800 500 300 200 100

=====PRODUCT PRICE SORTING SYSTEM=====
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Quick Sort
5. Display
6. Exit

Enter your choice: 6

Thank You!